

GATES Program Fact Sheet

The GATES program is a joint initiative between DOST, JICA DX Lab, and BCG to build a shared data and AI platform for disaster risk management in the Philippines. The platform's lakehouse runs on Apache Iceberg with Trino as the federated query engine, and dbt handles Bronze-to-Silver-to-Gold transformations.

The AI layer procured 24 H200 GPUs, arranged as three HGX nodes, to serve open-source large language models such as the Llama 3 family through vLLM and KServe. Retrieval-augmented generation uses Qdrant as the vector store, with OpenAI's large embedding model producing 1024-dimension vectors.

Two flagship agents run on this stack: a SQL query agent that answers natural-language questions against the lakehouse gold layer, and a PDRA agent that fuses PAGASA and PHIVOLCS hazard data with GeoRiskPH exposure data to draft pre-disaster risk narratives at the municipality level.

Observability is handled by Langfuse, tracking tokens, latency, and cost across both agent paths so the team can compare their operating profiles.